Model Validation Report

Trading Book 1-Day 99% Historical VaR

Item	Detail
Model ID / version	MR-VAR-004 / 1.4
Model type	Market risk — Value-at-Risk
Model owner	Market Risk Analytics
Validation run	20261023-194451-MR-VAR-004-9545cb
Automated analysis	2026-10-23 19:44 UTC (0.1 s)
Reviewed and signed off by	Demo reviewer, 2026-10-23 19:44 UTC
Overall rating	Unsatisfactory
Framework	Federal Reserve / OCC SR 11-7 — Supervisory Guidance on Model Risk Management

1. Executive summary

Overall rating: Unsatisfactory

The model should not be relied on for its intended use until High-severity findings are remediated, or only under compensating controls approved by the model risk committee.

The validation of Trading Book 1-Day 99% Historical VaR concludes with an overall rating of Unsatisfactory. The model should not be relied on for its intended use until High-severity findings are remediated, or only under compensating controls approved by the model risk committee. After review, 14 finding(s) were retained: 4 High, 7 Medium and 3 Low severity. The most material are Exception frequency inconsistent with the VaR confidence level (too many) (BT-001); Duplicate dates in price history (DQ-001); Zero or negative prices (DQ-002).

Executive summary drafted by template from the reviewed findings; ratings and severities are set by the validator.

Severity	Findings	Remediation due within
High	4	90 days
Medium	7	180 days
Low	3	365 days

Review status: 13 accepted, 1 edited, 0 rejected, 0 pending.

2. Model and validation scope

Purpose: Daily 1-day 99% Value-at-Risk for the multi-asset trading book.

Intended use: Daily risk limits monitoring and internal capital reporting.

Documented assumptions tested:

- The past 250 trading days are representative of next-day risk.
- Daily returns are independent across days.
- Portfolio weights are constant over the 1-day horizon.
- Price history is adjusted for corporate actions.

Documented limitations:

- Does not capture intraday liquidity risk.

Validation activities

Review area	SR 11-7 element	Status	Findings proposed
Data Quality	Evaluation of conceptual soundness	Completed	6
Assumptions	Evaluation of conceptual soundness	Completed	3
Backtesting	Outcomes analysis	Completed	3
Challenger Benchmarking	Ongoing monitoring	Completed	1
Sensitivity Analysis	Evaluation of conceptual soundness	Completed	1

Package validated (SHA-256 of each file, first 12 characters): model_package.yaml e64f07115ced; prices.csv bcdd0b7b521c

Data treatment applied before model-level tests: kept first of 1 duplicated date(s); treated 1 non-positive price(s) as missing; forward-filled 4 missing price(s).

3. Findings summary

ID	Severity	Finding	Area	Review	Due
BT-001	High	Exception frequency inconsistent with the VaR confidence level (too many)	Backtesting	Accepted	2027-01-21
DQ-001	High	Duplicate dates in price history	Data Quality	Accepted	2027-01-21
DQ-002	High	Zero or negative prices	Data Quality	Accepted	2027-01-21
DQ-003	High	Implausible one-day price moves	Data Quality	Accepted	2027-01-21
AS-001	Medium	Portfolio returns are serially correlated	Assumptions	Accepted	2027-04-21
AS-002	Medium	Volatility clustering in portfolio returns	Assumptions	Accepted	2027-04-21
AS-003	Medium	Recent volatility differs from the VaR window	Assumptions	Accepted	2027-04-21
BT-002	Medium	Basel traffic light in the yellow zone	Backtesting	Accepted	2027-04-21
CH-001	Medium	Volatility-sensitive challengers backtest better than the champion	Challenger Benchmarking	Accepted	2027-04-21
DQ-004	Medium	Gaps in price history	Data Quality	Accepted	2027-04-21
DQ-005	Medium	Stale (unchanged) prices	Data Quality	Accepted	2027-04-21
BT-003	Low	VaR exceptions cluster in time	Backtesting	Accepted	2027-10-23
DQ-006	Low	Missing prices	Data Quality	Edited	2027-10-23
SA-001	Low	Historical worst day far exceeds current VaR	Sensitivity Analysis	Accepted	2027-10-23

4. Detailed findings

Data Quality

The Data Quality review of Trading Book 1-Day 99% Historical VaR identified 6 finding(s) (3 high, 2 medium, 1 low severity). High-severity items: Duplicate dates in price history; Zero or negative prices; Implausible one-day price moves.

DQ-001 · Duplicate dates in price history

Severity: High · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-01-21

1 date(s) appear more than once (e.g. 2022-10-10), producing zero or double-counted returns.

Evidence

- duplicate_dates: [2022-10-10]
- count: 1

Recommendation: De-duplicate the time series and confirm which observation is the official close.

DQ-002 · Zero or negative prices

Severity: High · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-01-21

Non-positive prices found: {'RATES_10Y': 1}. Log returns are undefined and simple returns are -100%.

Evidence

- nonpositive_counts: {RATES_10Y: 1}

Recommendation: Replace with validated closes; if the series can legitimately be negative (e.g. spreads), model it in levels or changes, not returns.

DQ-003 · Implausible one-day price moves

Severity: High · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-01-21

One-day moves larger than 25% found, e.g. EQ_US -50.2% on 2022-12-19. A single move of this size usually signals an unadjusted corporate action (split, dividend) or a bad tick.

Evidence

- extreme_returns: {EQ_US: [{date: 2022-12-19, return: -0.502}]}
- threshold: 0.25

Recommendation: Verify against corporate-action records; use total-return adjusted prices. If the move is genuine, document it as a stress observation.

DQ-004 · Gaps in price history

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

1 gap(s) longer than 7 calendar days, e.g. 2023-07-14 to 2023-07-31. Returns spanning a gap overstate one-day moves.

Evidence

- gaps: [{from: 2023-07-14, to: 2023-07-31, days: 17}]
- threshold_days: 7

Recommendation: Back-fill the missing history from an alternate source or exclude returns that span gaps.

DQ-005 · Stale (unchanged) prices

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

Prices unchanged for 5+ consecutive observations: {'FX_EURUSD': {'runs': 1, 'longest_days': 10}}. Stale data understates volatility and VaR.

Evidence

- stale: {FX_EURUSD: {runs: 1, longest_days: 10}}
- threshold_days: 5

Recommendation: Check the market-data feed for the affected period and replace stale points with observed prices.

DQ-006 · Missing prices

Severity: Low · Evaluation of conceptual soundness · Review: edited · Remediation due 2027-10-23

Missing prices by asset (share of days): {'EQ_EU': 0.003}.

Evidence

- missing_rate: {EQ_EU: 0.003}

Recommendation: Document the forward-fill rule in the methodology.

Validator note (Demo reviewer): Three missing prices out of ~1,000; forward-fill impact is immaterial.

Assumptions

The Assumptions review of Trading Book 1-Day 99% Historical VaR identified 3 finding(s) (3 medium severity).

AS-001 · Portfolio returns are serially correlated

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

Ljung-Box test on returns (10 lags): Q = 38.1, p < 0.0001. The model assumes daily returns are independent.

Evidence

- q: 38.06
- p: 4e-05

Recommendation: Assess the impact on VaR, and on any square-root-of-time scaling to longer horizons.

AS-002 · Volatility clustering in portfolio returns

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

Ljung-Box test on squared returns: Q = 42.9, p < 0.0001. Volatility is persistent, so an equally weighted historical window reacts slowly when market conditions change.

Evidence

- q: 42.94
- p: 1e-05

Recommendation: Benchmark against a volatility-weighted (filtered) historical simulation or EWMA approach; monitor exception clustering in backtesting.

AS-003 · Recent volatility differs from the VaR window

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

Volatility over the last 60 days is 2.69x the rest of the 250-day window (higher). The documented assumption that the window represents next-day risk is weakened.

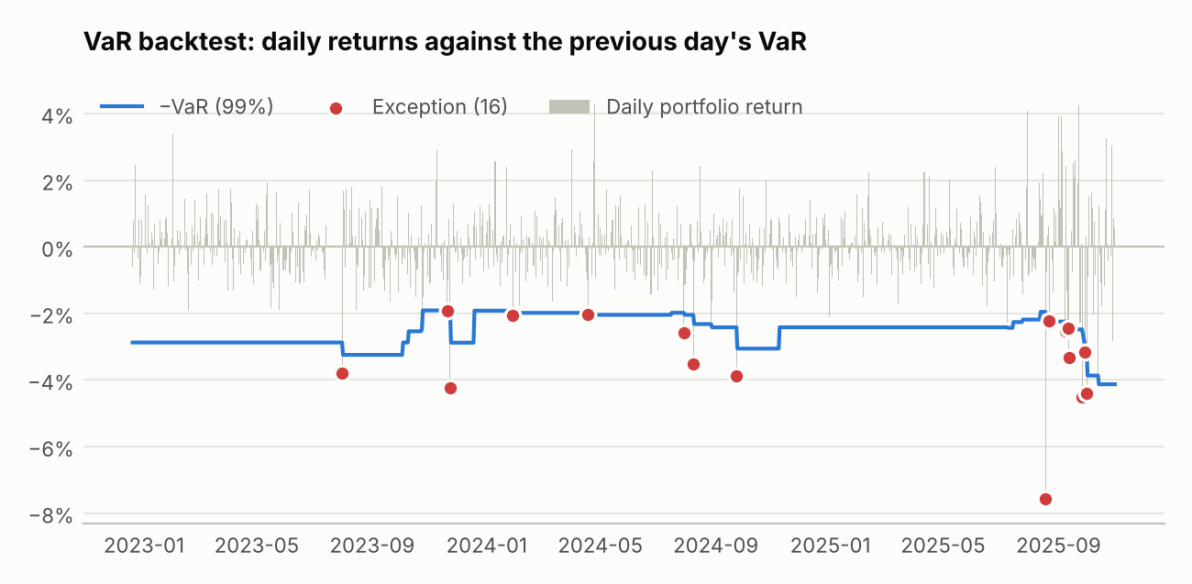
Evidence

- recent_vs_window_vol_ratio: 2.693
- recent_days: 60

Recommendation: Compare against a shorter or volatility-weighted window; if volatility is higher, VaR is likely understated until the window catches up.

Backtesting

The Backtesting review of Trading Book 1-Day 99% Historical VaR identified 3 finding(s) (1 high, 1 medium, 1 low severity). High-severity items: Exception frequency inconsistent with the VaR confidence level (too many).



Exceptions are days whose loss exceeded the VaR forecast made the previous day.

BT-001 · Exception frequency inconsistent with the VaR confidence level (too many)

Severity: High · Outcomes analysis · Review: accepted · Remediation due 2027-01-21

Kupiec POF test over 739 days: 16 exceptions vs 7.4 expected (LR = 7.60, p = 0.0058). VaR is understating risk.

Evidence

- exceptions: 16
- expected: 7.4
- lr: 7.6
- p: 0.005835

Recommendation: Re-examine the window length, weighting and data treatment; compare with a challenger VaR.

BT-002 · Basel traffic light in the yellow zone

Severity: Medium · Outcomes analysis · Review: accepted · Remediation due 2027-04-21

8 exceptions in the last 250 trading days against 2.5 expected at 99% confidence (cumulative binomial probability 99.8943%).

Evidence

- days: 250
- exceptions: 8
- zone: yellow

- cumulative_probability: 0.9989

Recommendation: Investigate each exception (model vs. P&L issue), escalate per the backtesting policy, and assess whether a capital multiplier add-on or model change is required.

BT-003 · VaR exceptions cluster in time

Severity: Low · Outcomes analysis · Review: accepted · Remediation due 2027-10-23

Christoffersen independence test: LR = 4.07, p = 0.044; 2 exception(s) followed another exception. The model is slow to react to changing volatility.

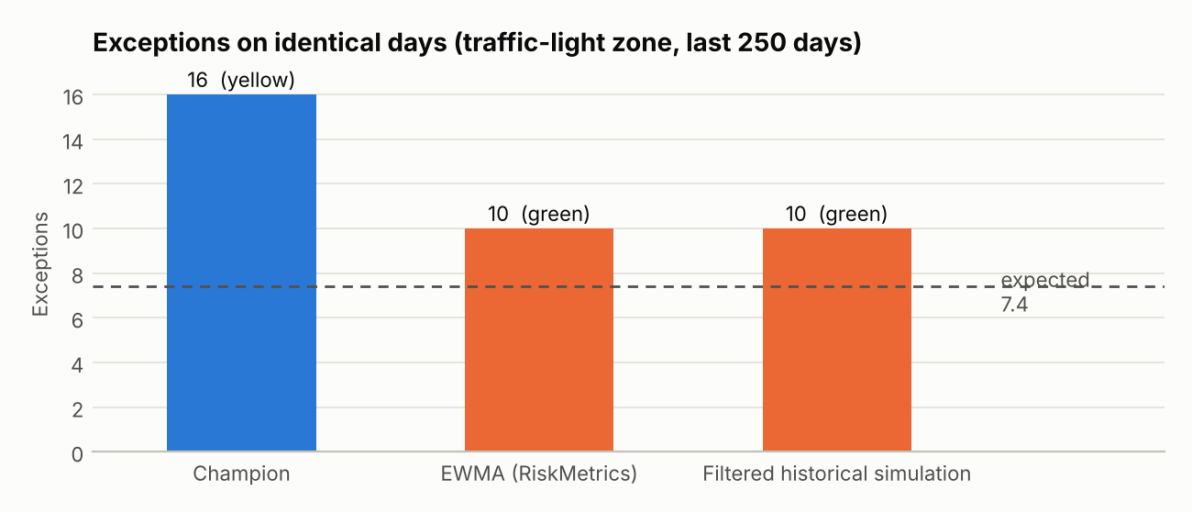
Evidence

- lr: 4.067
- p: 0.04372
- n00: 708
- n01: 14
- n10: 14
- n11: 2

Recommendation: Consider volatility scaling (EWMA / filtered historical simulation).

Challenger Benchmarking

The Challenger Benchmarking review of Trading Book 1-Day 99% Historical VaR identified 1 finding(s) (1 medium severity).



Exception counts over the same backtest days; dashed line = expected count.

CH-001 · Volatility-sensitive challengers backtest better than the champion

Severity: Medium · Ongoing monitoring · Review: accepted · Remediation due 2027-04-21

Over the same 739 days the champion has 16 exceptions (yellow zone), while EWMA (RiskMetrics) has 10 (green zone); Filtered historical simulation has 10 (green zone) against 7.4 expected. The challengers also have lower quantile loss.

Evidence

- Champion: {days: 739, exceptions: 16, expected: 7.4, kupiec_p: 0.00584, christoffersen_p: 0.04372, …}
- EWMA (RiskMetrics): {days: 739, exceptions: 10, expected: 7.4, kupiec_p: 0.3598, christoffersen_p: 0.6002, …}
- Filtered historical simulation: {days: 739, exceptions: 10, expected: 7.4, kupiec_p: 0.3598, christoffersen_p: 0.6002, …}

Recommendation: Evaluate adopting volatility scaling (EWMA / FHS) in the production VaR, or document why the equally weighted window is retained and how its lag is compensated.

Sensitivity Analysis

The Sensitivity Analysis review of Trading Book 1-Day 99% Historical VaR identified 1 finding(s) (1 low severity).

SA-001 · Historical worst day far exceeds current VaR

Severity: Low · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-10-23

The largest one-day loss in the history, 20.06% on 2022-12-19, is 4.9x the current VaR. Either the history contains a genuine tail event that VaR at this confidence level will not reflect, or the observation is a data error (check the Data Quality findings).

Evidence

- date: 2022-12-19
- loss: 0.2006
- multiple_of_var: 4.86

Recommendation: Confirm the observation; if genuine, cover it through stress testing alongside VaR.

Appendix A. Findings rejected by the validator

None.

Appendix B. Audit trail and turnaround

Measure	Value
Automated analysis	0.09 s
Data Quality	0.02 s
Assumptions	0.008 s
Backtesting	0.008 s
Challenger Benchmarking	0.026 s
Sensitivity Analysis	0.021 s
Review decisions recorded	14
Active review (first to last decision)	0.0 min
Run completion to sign-off	0.0 h
Total elapsed (start to sign-off)	0.0 h

Times are taken from the run's append-only audit log (audit.jsonl), which also records every review decision with reviewer, timestamp and rationale.

Sign-off comment: Illustrative review of a synthetic model package.

Appendix C. Limitations of this validation

- Automated tests cover the checks listed in section 2; they do not replace the validator's judgement on conceptual soundness, business use or the appropriateness of the model's design.

- Statistical tests reject a correct model at roughly their significance level; findings are therefore reviewed individually before they are accepted.
- AI-written text (area summaries, commentary, executive summary) is generated from the computed results only and is labelled as such; it cannot add, remove or re-grade findings.